Once breakpoints are set up, we can issue the run command to gdb to run the application code inside the debugger. This runs the code up until the first breakpoint is encountered. At this point, we can step through the code while examining how the value of different variables change. We can step through the code source one line at a time, or even at the assembly level, one instruction at a time. We can inspect the contents of the memory and even low level registers. And then even modify memory locations, if needed.

Let's look at how to do most of these things.

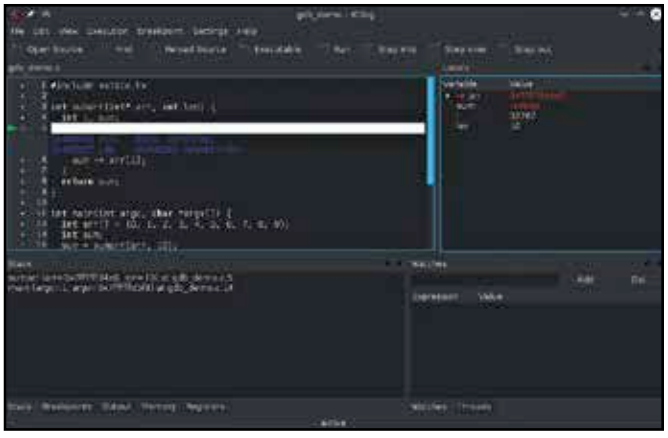
Example code

Before we start the debugging code, let's take a look at the sample buggy code we'll be using. The bugs in this code may be obvious, but remember that probably wont be the case when you're using a debugger.

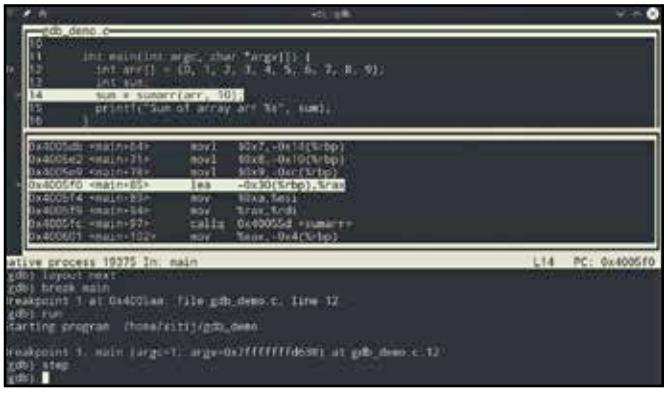
```
#include <stdio.h>

int sumarr(int* arr, int len) {
 int i, sum;
 for (i = 0; i <= len; ++i) {
  sum += arr[i];
 }
 return sum;
}

int main(int argc, char *argv[]) {
 int arr[] = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9};
 int *sum;
 *sum = sumarr(arr, 10);
 printf("Sum of array arr %s", sum);
}
```



Kdbg is a GUI frontend for gdb developed by KDE. It is quite powerful and includes most of the features you'd need for debugging



gdb includes a curses-based UI that might be more comfortable for some. You launch gdb in this mode with gdb -tui

Running and viewing code

Once you've started gdb and provided it the path to your application, you can run the application by simply typing "run". This will run the program as normal, producing output, asking for input etc. You can also provide command line arguments (if any) to the run command.

If the program crashes while running under gdb, you'll notice that you get a lot more information than you would if you were running it directly. For example, it will tell you where exactly in your program (or out of it) the crash occurred, and will even display the code for that line.

If you want even more information, you can type the "backtrace" command to get the exact chain of function calls that led to the crash.

At this point if you want you can type list to list the code around this area. You can even provide list a line number and it will print the code around that line.

It can sometimes also be useful to see the assembly code generated from your source code, if the problem isn't visible at a high level. To view the assembly code for your program type disas. With disas /m you can see your own code interleaved with the assembly code generated from it. It's also a brilliant way to get familiar with assembly.

Chances are you want to pause running your application before it reaches the point that leads to a crash so you can diagnose the state of the application there. For that you need to set breakpoints, we look at those next.

Setting breakpoints

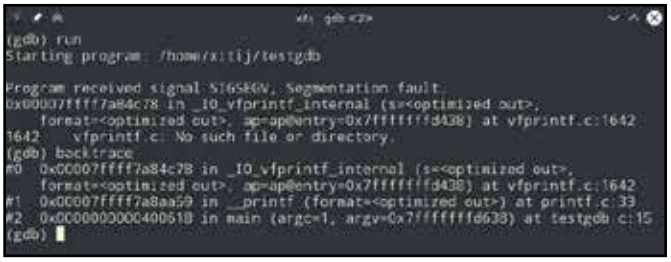
A breakpoint is place within your code where gdb will halt the automatic execution of code and turn over control to you.

There are a number of ways that you can set breakpoints. The simplest way to set a breakpoint is to specify a line number at which the code should stop executing. You can use the list command explained above to figure out what the line numbers are.

Another way to set a breakpoint is to set it using the name of the function. In this case execution halts if that function is called. Withbreak main you can halt execution when your program starts executing its main function.

A more interesting way to set a breakpoint, is to have a conditional break based on the state of the program. For instance you might want to halt the code if a particular variable reaches a certain value. You can specify such a break as follows:

```
break if i > 5
```



If an application keeps crashing on launch sometimes running it under gdb can give you a hint as to why.